

VP OF AI · 4MP, A XORA
PORTFOLIO COMPANY

Autonomy begins when a correction has earned the right to act.

Dear 4MP Founding Leadership Team,

4MP's problem is compelling because it starts where digital perfection meets physical reality. CAD and CAM encode intent; machines, tools, fixtures, materials, temperature, operators, and time introduce variation. The opportunity is not simply to detect that gap. It is to build an intelligence layer that can explain the difference, propose a bounded response, verify the result, and preserve only the learning that is safe to reuse.

That integration point has defined much of my career. At Vape-Jet, I connect factory, field, customer, machine-fleet, quality, support, engineering, ERP, and telemetry signals in a software-enabled robotics and machine-vision environment with approximately 1,000 fielded robots. At Compunetics, I led engineering, manufacturing, quality, and high-reliability technical programs while building root-cause practices, process controls, and standard work. Through ZeusVu and DudeWorth, I have worked across regulated autonomous systems, TensorFlow-based computer-vision concepts, agentic workflows, human authority, evaluation, escalation, and governance.

For 4MP, I would organize the AI architecture around the correction itself: manufacturing intent, reconstructed physical state, cause hypotheses, a proposed action, explicit authority, validation evidence, observed outcome, and a defined boundary for reuse. That creates a practical bridge among G-code and CAM context, geometry, metrology, physics-based models, machine telemetry, simulation, operators, and a scalable data platform. It also creates a disciplined path from shadow recommendation to bounded autonomy.

The first 120 days should prove one valuable correction class end to end while installing the system that allows many more to follow. I would begin with a shared correction object and evidence contract, close a narrow shadow loop, stress its boundary across real variation, and translate the proof into team design, simulation and data architecture, technical hiring priorities, release gates, and a correction-class roadmap.

I built an independent candidate vision at <https://russelldudek.github.io/Xora/>. It includes the interactive Qualification Lens, the 120-day entry plan, and a Correction Qualification Record that makes my reasoning visible without claiming knowledge of 4MP's undisclosed architecture.

I would welcome the opportunity to put one representative correction problem on the table with your software, machine-physics, geometry, metrology, and data leaders and work it through from intent to qualified learning.

Carpe diem,

Russell Dudek

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How I would contribute

Build one coherent system across three inseparable layers:

- **Intelligence:** interpret intent, reconstruct state, rank causes, and propose bounded action.
- **Evidence:** replay, simulate, test, measure, and define where learning is valid.
- **Organization:** recruit technical depth, clarify ownership, and install correction-review discipline.

Working thesis

Make every correction explainable, bounded, validated, and reusable.